impl Sample for ZExpFamily<1>

Michael Shoemate (@Shoeboxam)

This proof resides in **"contrib"** because it has not completed the vetting process.

This proof relies on caller-verified preconditions, so the corresponding API belongs behind the **"honest-but-curious"** feature flag.

Proves soundness of the implementation of `Sample` for `ZExpFamily<1>` in `mod.rs` at commit f5bb719 (outdated[1]).

1 Hoare Triple

Precondition

Compiler-verified

None

Caller-verified

`self` represents a valid probability distribution.

Pseudocode

```
1 class ZExpFamily1: # analogous to impl Sample for ZExpFamily<1> in Rust
2     def sample(self, shift):
3         return shift + sample_discrete_laplace(self.scale) #
```

Postcondition

Theorem 1.1. Either returns `Err(e)` independently of the input `shift`, or `Ok(shift + Z)` where `Z` is a sample from the distribution defined by `self`.

Proof. By the precondition, since `self` represents a valid probability distribution, then by the definition of `ZExpFamily`, `self.scale` is non-negative.

Since the preconditions for are met (non-negative scale), then we claim the postcondition that the returned value follows a given distribution. This distribution matches that defined by `self`.

By the addition on line 3, the implementation satisfies the postcondition. □

[1] See new changes with `git diff f5bb719..58d6f416 rust/src/measurements/noise/mod.rs`